

2. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX 100

int main()
{
    char input[MAX];
    int i;

    while (1)
    {
        printf("myshell> ");

        i = 0;

        while (1)
        {
            char ch = getchar();

            /* Enter key */
            if (ch == '\n')
            {
                input[i] = '\0';
                printf("\n");
                break;
            }

            /* Backspace */
            else if (ch == '\b' || ch == 127)
            {
                if (i > 0)
                {
                    i--;
                    printf("\b \b");
                    fflush(stdout);
                }
            }

            /* Store normal characters */
            else if (i < MAX - 1)
            {
                input[i] = ch;
            }
        }
    }
}
```

```
GNU nano 8.7.1 skill2.
        else if (i < MAX - 1)
        {
            input[i] = ch;
            i++;
            putchar(ch);
            fflush(stdout);
        }
    }

    /* Exit condition */
    if (strcmp(input, "exit") == 0)
    {
        printf("Exiting shell...\n");
        break;
    }

    /* Empty input */
    if (strlen(input) == 0)
    {
        continue;
    }

    /* Example commands */
    if (strcmp(input, "hello") == 0)
    {
        printf("Hello! Welcome to my shell.\n");
    }
    else if (strcmp(input, "help") == 0)
    {
        printf("Available commands:\n");
        printf("hello\n");
        printf("help\n");
        printf("exit\n");
    }
    else
    {
        printf("Command not found: %s\n", input);
    }
}

return 0;
}
```

```
-rw-r--r-- 1 thumu_hemanth_kumar thumu_hemanth_kumar 188 Aug 14 08:53 skill1b.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ nano skill2.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ gcc skill2.c -o skill2
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill2
myshell> ^C
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill2
myshell> hello
hello
Command not found: hello
myshell> help
help
Available commands:
hello
help
exit
myshell>

myshell> helloo
helloo
Command not found: helloo
myshell> hello
hello
Hello! Welcome to my shell.
myshell> exit
exit
Exiting shell...
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ |
```